Lemma 2.41. *Let $U \subseteq \mathrm{RV}^k$ be a doubly bounded set and $f : U \longrightarrow \mathrm{RV}$ a definable function. Then $f(U)$ is also doubly bounded. The same holds if the codomain of f is Γ.*

Proof. For an RV-valued f, put $h = \mathrm{vrv} \circ f$; for a Γ-valued f, put $h = f$. We retain the induction on k, with the case $k = 0$ immediate. For $k > 1$ and $t \in \mathrm{pr}_1(U)$, the inductive hypothesis makes $h(U_t)$ doubly bounded. Its finite bound

$$\gamma_t = \sup\{|h(t, x)| \mid x \in U_t\}$$

exists by definable completeness of Γ, and the least-upper-bound formula makes $t \longmapsto \gamma_t$ definable. The projection $\mathrm{pr}_1(U)$ is doubly bounded, so the unary Γ-valued case bounds all γ_t and hence bounds $h(U)$. Thus both claims reduce to the case $k = 1$ and a definable function $g : U \longrightarrow \Gamma$. Then, by C-minimality, we may assume that g vrv-contracts to a function $g_\downarrow : \mathrm{vrv}(U) \longrightarrow \Gamma$. Since definable functions in Γ are piecewise $\mathbb{Q}$-linear, the range of $g_\downarrow$ must be doubly bounded. □

Resolved referee R037 [Gap] – Uniformity in the bounded RV-image lemma.
Original source line 1017. The induction on k must control the bounds uniformly in the remaining coordinates; boundedness on each fiber alone does not give a bound on their union. A precise route is to consider the relation of possible values $\mathrm{vrv}(f(u))$ over $\mathrm{vrv}(u) \in \Gamma^k$. Residue-field/value-group orthogonality makes each such fiber finite, uniformly; stable embeddedness and elimination in divisible ordered abelian groups then give finitely many piecewise rational-affine branches. These have bounded images on a bounded box. Include this justification, or an equivalent explicit uniform induction.

Author resolution. The author retains the original induction. Its fiber bounds are chosen definably as finite suprema of the absolute output values; the unary Γ-valued case then bounds this definable bound function uniformly. The proof now spells out that step, without replacing the induction by a relation of all input and output values.

Definition 2.42. Let $A \subseteq \mathrm{VF}^n$ and $f : A \longrightarrow \mathcal{P}(\mathrm{VF}^m)$ be a definable function. For $a \in \mathrm{VF}^n$ and $L \subseteq \mathrm{VF}^m$, we say that L is a *limit set of f at a*, written as $\lim_{A \to a} f \subseteq L$, if for every $\epsilon \in \Gamma$ there is a $\delta \in \Gamma$ such that if $c \in \mathfrak{o}(a, \delta) \cap (A \setminus \{a\})$ then

$$f(c) \subseteq \bigcup_{b \in L} \mathfrak{o}(b, \epsilon).$$

This is a local definition and imposes no global boundedness hypothesis. A finite limit set forces the union of the values $f(c)$ to be bounded on some punctured neighborhood of a. Whenever boundedness of the range is assumed, it means boundedness of $\bigcup_{c \in A} f(c)$.

A limit set L of f at a is *minimal* if no proper subset of L is a limit set of f at a.

Resolved referee R038 [Clarification; revised] – Simplify the limit-set definition.
Original source line 1046. Say explicitly that $\bigcup_{a \in A} f(a)$ is bounded. The auxiliary subset L' does no work: the displayed condition is equivalent to $f(c) \subseteq \bigcup_{b \in L} \mathfrak{o}(b, \epsilon)$. The following definition of continuity is applied to finitary functions without a global boundedness hypothesis, so the present definition needs a compatible extension. One can use the same formula for arbitrary correspondences whenever a finite affine limit exists, or formulate limits in $(\mathbb{P}^1(\mathrm{VF}))^m$ from the outset; local boundedness near the point suffices for affine local uses.

Author resolution. The author approves the local limit formula without a global range bound, using all of L in place of the redundant subfamily L'. A finite limit set supplies local boundedness on a punctured neighborhood. Any separately assumed range bound refers to the union of all values. Minimality and the puncture are unchanged.